

# SUHAIL

## DATA SCIENTIST | APPLIED MACHINE LEARNING

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### PROFESSIONAL SUMMARY

Data Scientist with 4+ years of experience across data analytics, healthcare, and technology-driven environments, specializing in Python, SQL, predictive modeling, and applied machine learning. Experienced in transforming complex data into analytical and ML solutions, with hands-on work spanning feature engineering, model evaluation, data quality, and reproducible ML workflows. Combines data science expertise with PostgreSQL, REST API, AWS, and production-system experience to build ML solutions designed for real-world integration and deployment.

### TECHNICAL SKILLS

**Programming:** Python, SQL, JavaScript

**Machine Learning & Data Science:** pandas, NumPy, scikit-learn, XGBoost, SciPy, statsmodels, SHAP, feature engineering, predictive modeling, time-series forecasting, model evaluation

**MLOps & ML Engineering:** MLflow, FastAPI, Docker, GitHub Actions, experiment tracking, model versioning, model serving, monitoring

**Data & Visualization:** PostgreSQL, Redis, Excel, Power BI, Tableau, Matplotlib

**Cloud & Production:** AWS (EC2, S3, RDS, IAM), REST APIs, Node.js, Express.js, Nginx, Linux, Git/GitHub

### PROFESSIONAL EXPERIENCE

#### NYRX at Ralph LLC

New York, NY

##### Data Scientist

Mar 2026 - Present

- Developed a medication demand forecasting solution from pharmacy billing and ordering history, using time-based features and predictive models to estimate future demand and surface potential inventory mismatches.
- Established the ML workflow from data preparation through model evaluation and deployment, incorporating experiment tracking, model versioning, containerized inference, and performance monitoring for repeatable model releases.
- Designed PostgreSQL data models and validation workflows for healthcare and pharmacy data, improving the reliability of data used across analytics, operational reporting, and machine-learning workflows.
- Built and maintained REST APIs and backend services with Node.js, Express.js, PostgreSQL, Redis, and AWS, providing the production infrastructure needed to connect application workflows with analytical and ML components.
- Led backend and data implementation within a five-person engineering team, working directly with company leadership and coordinating with frontend, DevOps, and QA through development, testing, and deployment.

##### Junior Data Analyst

Oct 2025 - Mar 2026

- Analyzed pharmacy billing, inventory, and operational data using SQL, Excel, and Power BI, cleaning and reconciling records to identify quantity discrepancies, usage patterns, and inventory trends.
- Built reporting workflows and dashboards to compare medication billing and ordering activity, giving management clearer visibility into shortages, excess quantities, and purchasing needs.

#### Hands Industries FZC

Sharjah, UAE (Remote)

##### Data Analyst I

Aug 2020 - Oct 2023

- Analyzed customer, sales, and operational datasets using SQL, Excel, Tableau, and Power BI, cleaning and organizing approximately 5,000-10,000 records to identify customer trends, recurring issues, and business performance patterns.
- Built dashboards and structured reporting workflows from manually maintained business records, helping management track customer and operational trends and use data more effectively for decision-making.

### PROJECTS

#### 30-Day Hospital Readmission Risk Prediction

Python | pandas | scikit-learn | XGBoost | CatBoost | SHAP

- Built an end-to-end ML pipeline on **100K+ diabetic hospital encounters**, with patient-level leakage prevention, ICD-9 feature engineering, class-imbalance handling, and grouped 5-fold cross-validation.
- Tuned and selected **XGBoost**, achieving **0.232 PR-AUC**, **0.680 ROC-AUC**, and **71.2% recall** on a locked **10K-encounter test set**, with validation-based threshold selection for readmission-risk prioritization.
- Applied **TreeSHAP and error analysis** to explain prediction drivers and investigate false positives, false negatives, and subgroup performance disparities.

#### Pharmacy Reconciliation & Prescription Renewal ML Platform

Python | scikit-learn | MLflow | FastAPI | PostgreSQL | Docker

- Built a leakage-controlled prescription-renewal prediction pipeline on synthetic longitudinal pharmacy data, engineering point-in-time features and using chronological validation, time-series cross-validation, and model comparison to lock a regularized Logistic Regression model before final testing.
- Operationalized the ML workflow with MLflow experiment tracking and model versioning, FastAPI inference, PostgreSQL/Alembic persistence, Docker Compose, health monitoring, privacy-safe request logging, automated testing, and GitHub Actions CI.

### RESEARCH & TECH. REPORTS

#### MedTutorDx — LLM Medical Reasoning & Evaluation Benchmark

Manuscript in Preparation

- Conducting controlled experiments on medical AI reasoning and question-answering across multiple medical subject areas using reproducible evaluation criteria and statistical analysis.

#### A Desktop-Based Analytics System for Pharmacy Billing and Order Reconciliation

2026 | Technical Report | DOI: 10.5281/zenodo.18652148

- Designed and evaluated an applied pharmacy analytics system for billing and order reconciliation, supporting scalable filtering, discrepancy analysis, and operational decision support across 700+ medications.

### EDUCATION

#### St. Francis College | Brooklyn, NY

Master of Science in Information Technology Management | GPA: 3.85/4.00 | Aug 2025

#### Osmania University | Hyderabad, India

Bachelor of Engineering in Computer Science & Engineering | GPA: 3.67/4.00 | Jul 2023

### LEADERSHIP

#### President, Ace Lords - Department of Computer Science & Engineering | L.I.E.T | 2021-2023

- Led a 15-member technical organization serving a 300-400 student CSE department; organized 10 technical workshops, a Datathon, and a three-day technical event featuring 8 competitions and 800+ participants. Received a Letter of Appreciation for two years of leadership.